

A Mars Starship Design for Humans

Vehicle Design - Blue Team

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16 March 2025

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Abstract—Human space exploration is advancing toward Mars, with SpaceX’s Starship envisioned as the primary vehicle for transporting crews to the red planet. This project focuses on designing a Mars-bound Starship optimized for human spaceflight, addressing key challenges such as long-duration life support, radiation shielding, and spacecraft habitability. The Vehicle Design subgroup is responsible for modifying the payload section of Starship to accommodate six crew members while maintaining mass efficiency, structural integrity, and operational reliability. Our design includes crew living and working areas, power generation and management, thermal regulation, data handling and communication systems. Given the harsh environment of deep space and the Martian surface, we integrate solutions for radiation protection, redundancy in critical systems, and sustainable resource utilization. The project assumes an unmanned Starship mission in 2028 to pre-deploy infrastructure for in-situ resource utilization, enabling propellant production for the return journey. While Starship’s exterior and propulsion systems remain unchanged, we focus on optimizing the interior layout to ensure astronaut safety and efficiency in microgravity and partial gravity environments. By addressing these engineering challenges, this project contributes to the broader effort of enabling sustainable Mars missions. Our work provides valuable insights into the design adaptations needed for deep-space travel, supporting the long-term goal of making human settlement on Mars a reality.

Index Terms—Human spaceflight, Mars mission, SpaceX Starship, Vehicle design, Life support systems, Radiation shielding, Deep-space travel, Thermal regulation, Power management, Crew habitation, ISRU.

I. INTRODUCTION

Human space exploration is on the brink of a new era, with Mars as the next frontier. SpaceX’s Starship, a fully reusable spacecraft, is designed to enable large-scale human missions beyond Earth’s orbit. This project focuses on designing a Mars-bound Starship capable of carrying and sustaining a human crew for a round-trip mission. Given the extreme challenges of interplanetary travel, our work emphasizes adapting Starship’s design to support human life for extended durations, ensuring structural integrity, efficient propulsion, and a robust life support system. Key design considerations include optimizing internal space for crew functionality, assessing power requirements, and integrating radiation shielding for deep-space travel. Additionally, we evaluate modifications needed to accommodate the challenges of Mars surface operations, assuming that an unmanned Starship will launch in 2028 to prepare for the first human mission in 2030. By addressing these design challenges, this work

contributes to the broader vision of sustainable human presence on Mars.

This project is part of the Blue Team, which is tasked with developing a comprehensive plan to transport humans to Mars and back, starting in 2030. The Blue Team is divided into five subgroups: Flight Design, Human Aspects, Life Support and Thermal Systems, Overall Coordination, and Vehicle Design. Our subgroup, Vehicle Design, focuses on the conceptual design of Starship’s interior, ensuring that all necessary components for human survival are integrated. This includes heat and cooling systems, communication, and power management. While the outer design and propulsion system of Starship remain mostly unchanged, the payload area is reconfigured to create a livable space for six crew members and their cargo. Through this work, we aim to develop a functional and sustainable spacecraft interior that meets the demands of deep-space travel and Mars surface operations.



Fig. 1: Mission Patch for Mars Starship Design

Figure 1 shows the mission patch for the Vehicle Design team of Blue Team. It showcases the mission name PRIME (Pioneering Research Innovation for Mars Exploration) along with the total mission duration of 969 Martian sols. The patch features two vehicles, Hugin and Munin, representing the transfer and elevator ships, respectively. Additionally, it depicts a Martian habitat,

which is the ultimate goal of the mission, along with the desired flight plan of the two ships. The patch also highlights that the mission will include a crew of six astronauts.

II. RELEVANT ASSUMPTIONS AND SYSTEM REQUIREMENTS

For the vehicle design of this mission, the following assumptions are given from discussions with the team. The design accounts for all phases of the crewed mission, starting from launch on Earth and ending with landing back on Earth. Earlier missions to bring supplies to Mars are not designed in detail, with considerations only being made regarding what cargo will be sent in advance. It is assumed that the earlier supply mission in 2028 allow us to only bring the food, water and other life support supplies needed for the transfer to Mars onboard the crewed mission. Additionally, recycling systems, nuclear reactor for power generation with solar & battery backup and a rover, necessary to commute on the martian terrain for other mission scenarios, are made available this way.

To fit Δv -requirements, the mission will comprise two Starships which are human rated; a transfer vehicle (TV) for going from Earth to Mars and back again, and a landing vehicle (LV) already situated in a Mars orbit for landing and takeoff on and from Mars. Both Starships are similar in design; any relevant differences are stated as such in this report. The crew is made up off 6 astronauts, all of which are accounted for in all phases of the mission (including Mars landing). The total mission duration is 996 days, of which 285 days are the transfer to Mars, 495 days are the stay on Mars' surface, and 216 days are the transfer back to Earth.

SpaceX officially states a 100 ton payload mass budget for a Mars mission using Starship. Since our design removes the fins of the Starship (25 t), a 100 ton payload requirement leaves us with a 20% margin. This budget includes everything designed within the scope of this project.

From these assumptions, requirements to the overarching system were defined, which can be found in Table I. From these, subsystem level (L2) requirements were defined and then used to design the vehicle; details on this can be found in the following chapters.

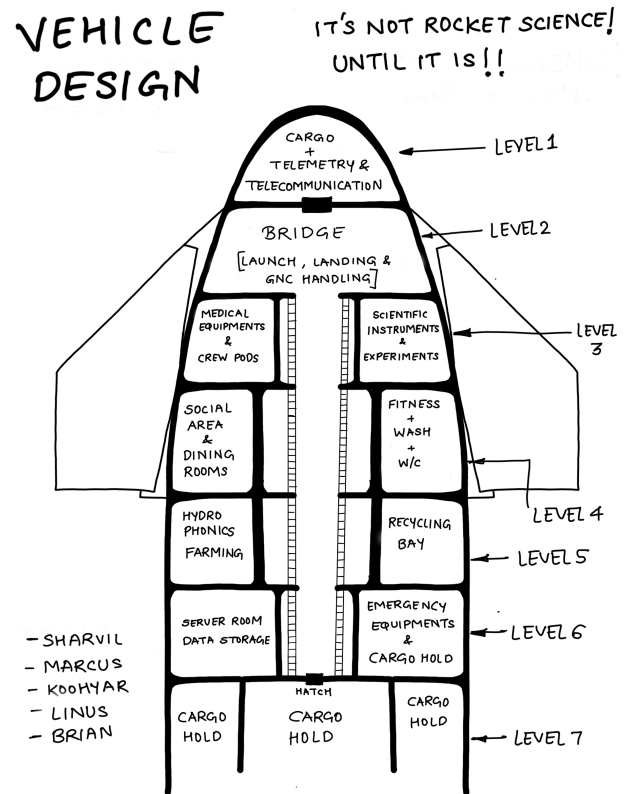
TABLE I: System (L1) Requirements

ID	Requirement Description
SYS-001	The system shall be able to facilitate a round trip mission to and from Mars, including a required stay on Mars' surface.
SYS-002	The system shall be able to accommodate a crew of 6 astronauts.
SYS-003	The system shall include necessary technologies to ensure the astronauts' well-being and success of the mission, including supply of power, thermal regulation, shielding and communication with ground stations.
SYS-004	The payload mass shall not exceed 100 tons.

III. VEHICLE OVERVIEW AND DESIGN

The design was initiated by creating a two-dimensional schematic of the payload area of Starship. This schematic was created to get a first look and idea of how the flooring and division of components inside the Starship would look. The schematic includes all the necessary components and material needed for the Mars mission. The schematic was revised and improved during the progression of the project with the input of other groups to fit all the requirements. The first and final version of the schematic can be seen in Figure 2 & 3.

By comparing the two schematics in Figure 2 & 3



TEAM BLUE

Fig. 2: Preliminary Iteration

one can note multiple changes. First of all, the steering fins for landing have been removed, this is because there will be a second Starship in orbit of Mars that will work as a transport between Mars orbit and the surface of Mars. This allows for more payload which was needed for accommodating all the radiation shielding necessary for the health of the crew. Another change made is the combination of floor 4, 5 and 6 into one living area while some equipment and storage have been placed around walls instead of on separate floors. This is also done to reduce the total weight of the radiation shielding by putting the main shielding on the inner walls to lower the total amount of surface area. Furthermore, an RCS system is added for maneuvering the spacecraft in orbit to point in any specific direction for communications or power, etc.

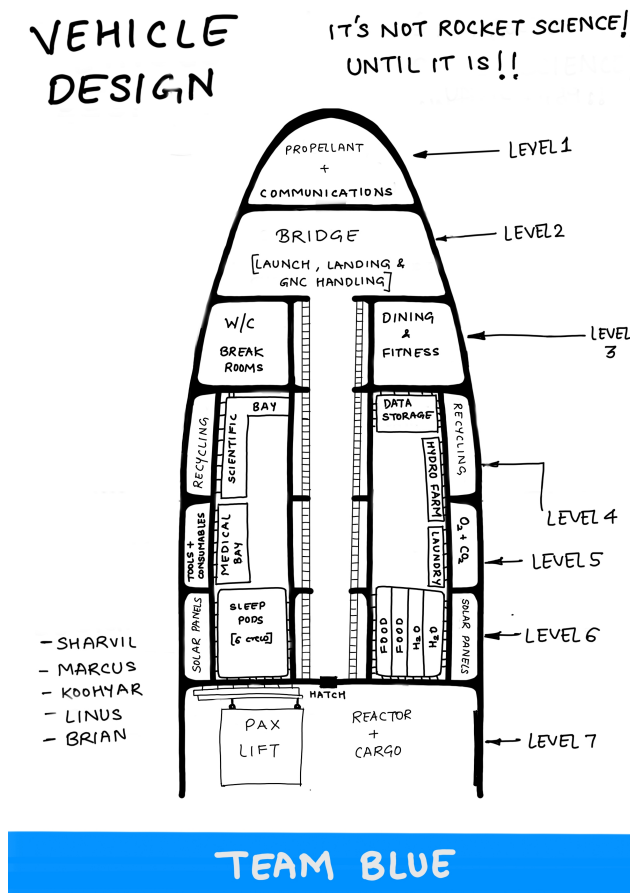


Fig. 3: Final Iteration

After the first schematic was made, the three-dimensional design was started. To create the three-dimensional design, a computer-aided design (CAD) program (Fusion 360) was used. First of all the shell and propulsion system was designed to look identical to SpaceX Starship, in order to get all the correct dimensions for the payload area where all storage and living area would be. The CAD design of Starship can be seen in Figure 4. A detailed description of location of all the subsystems is available in the appendix section at the end of this document.

Sub-Systems

All sub-systems in the schematic have been distributed around the interior of the Starship. These sub-systems have been designed and configured to satisfy the needs of the crew. Figures of most subsystems can be found in the appendix.

1) *Lift & EVA bay:* The lift system is included in the final schematic for the Earth-to-Mars transfer vehicle, even though it is primarily required for the landing Starship. Its placement in the schematic serves to illustrate its position within the Starship design. The lift is specifically designed to transport crew and cargo between the spacecraft and the Martian surface, ensuring a safe and efficient transition during Mars surface operations and eventual ascent back to the vehicle.

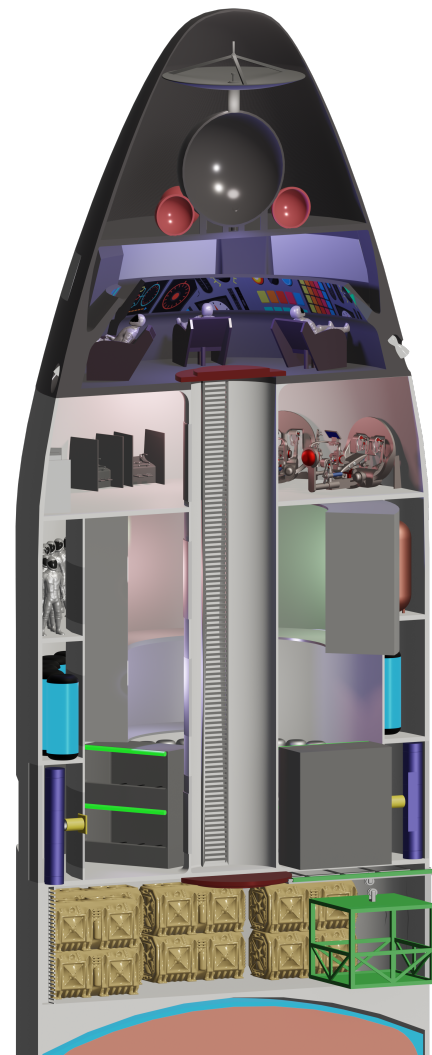


Fig. 4: Starship CAD - Payload section

Additionally, the bottom section of the habitable zone can function as an EVA (Extravehicular Activity) zone when required[1]. This area will be equipped with airlock mechanisms, suit donning stations, and storage for EVA tools. This dual-purpose design maximizes space utilization and mission flexibility, ensuring that the Starship can efficiently support both surface operations and deep-space travel.

2) *Sleeping Capsule:* The sleeping capsules are designed to provide each and every crew member with some personal space. In a capsule, a crew member can both lay down and sit. To prevent the person from floating away, straps are installed, which will simulate laying down in a normal bed. They also have a cover that can be rolled down and attached with velcro straps.

3) *Exercise machine:* Maintaining astronaut health and fitness is crucial for long-duration space missions, as prolonged exposure to microgravity leads to muscle atrophy and bone density loss. Instead of using multiple exercise devices, as seen on the ISS, our approach is to develop

a single, multifunctional exercise machine allowing astronauts to perform a wide range of exercises necessary for maintaining physical health.

To ensure redundancy and accessibility, two identical machines will be installed onboard the Mars Starship. The design, based on the exercise machine concept by J. Chang [2], will be optimized for both microgravity operations during transit and Martian gravity on the surface. By consolidating multiple fitness systems into a single, adaptable unit, we aim to maximize efficiency, reliability, and crew well-being throughout the 969-sol mission.

4) *WC*: The toilets are designed to work similarly to the ones on the international space station (ISS). They have a tube for urination (with their individual urination funnel) and a hole for defecation. They also have handles which can be placed over the astronauts thighs to keep them in a stable position, as well as straps to keep the feet in place.

5) *Guidance RCS*: The Mars Starship's guidance and control system relies on 12 RCS thrusters mounted in a circular configuration to provide precise attitude control and maneuverability. These thrusters enable yaw, pitch, and roll adjustments, as well as translational movements, ensuring stability during interplanetary travel, docking, and entry, descent, and landing (EDL). The system operates autonomously, using inertial measurement units (IMUs), gyroscopes, and star trackers to continuously adjust the spacecraft's orientation and trajectory. During spaceflight, the RCS system maintains optimal solar panel alignment and heat dissipation, while in EDL, it helps stabilize the vehicle for controlled atmospheric entry. Powered by hypergolic propellants for instant ignition and reliability, the system is designed with redundancy to ensure continued functionality even in case of individual thruster failure. This robust and efficient RCS network guarantees precise maneuverability across all mission phases, from launch to landing and return.

6) *Recycling System*: For a 969-sol Mars mission, efficient recycling systems are essential to minimize resource consumption and ensure crew sustainability. The Mars Starship integrates closed-loop recycling technologies for water, air, waste, and food production, reducing dependency on resupply missions from Earth.

The water recycling system collects and purifies urine, sweat, and condensation using multi-stage filtration, vapor compression distillation, and reverse osmosis. This ensures that nearly 98% of water is reclaimed and made potable. The air revitalization system includes CO scrubbers using liquid amine and solid-state filters to remove carbon dioxide, while electrolysis units generate oxygen by splitting water molecules. To enhance sustainability, microalgae-based bioreactors provide supplemental oxygen production and CO absorption.

Waste is managed through an advanced solid waste recycling system, where organic material is processed into fertilizer for hydroponic plant growth, and non-organic waste is compacted or repurposed. The nutrient recycling

system converts metabolic byproducts into usable compounds, supporting closed-loop agriculture for fresh food production. Additionally, thermal and power recycling systems maximize efficiency by using heat exchangers and regenerative fuel cells to optimize energy use.

By integrating these high-efficiency recycling systems, the Mars Starship ensures that air, water, and essential nutrients are continuously regenerated, enabling sustained human presence on Mars with minimal resource waste. The resulting L2 requirements are found in the appendix.

IV. STRUCTURE

The structural design of the Mars Starship is based on the existing cargo variant of SpaceX's Starship but has been significantly modified to accommodate a human-rated interior [3]. Unlike its original configuration, which prioritizes payload capacity, this version is optimized for long-duration space travel, ensuring crew safety, comfort, and functionality during the interplanetary journey and surface operations.

One of the key structural changes involves reconfiguring the cargo bay into a livable environment. The interior is divided into multiple sections, including crew quarters, a communal living area, a laboratory, exercise facilities, and a life support hub. These compartments are designed using modular elements, allowing flexibility in layout while ensuring efficient use of available space. Microgravity adaptation features, such as handrails, footholds, and reconfigurable workstations, have been integrated to facilitate movement and daily activities.

The pressure vessel has been reinforced to withstand deep-space conditions, providing enhanced radiation shielding using lightweight yet durable materials, such as polyethylene composites and water-lined walls in key areas. Additional thermal protection measures have been implemented to regulate internal temperatures against the extreme variations encountered during transit and surface stays.

To support crew safety, redundant structural reinforcements ensure durability under varying stress conditions, including launch, atmospheric entry, and landing. The vehicle's hull design remains largely unchanged, preserving Starship's aerodynamic profile, but the landing legs and structural supports have been improved for repeated landings on both Mars and Earth. By adapting the Starship's structural design to prioritize human habitation, this version ensures that astronauts can safely complete their mission while maintaining operational efficiency, sustainability, and long-term habitability.

Since there are two starships, one for transfer and the other for landing on Mars, both will have same structural components inside them as it is dependent on the usage. The landing vehicle is considered as the one which is already in orbit and will have mainly supplies instead of research and sleeping bays which are on the transfer starship. A detailed breakdown of the inclusions and exclusions in each of the starship is mentioned in the table below. This is in relation to the 2D design shown in the Vehicle Overview section. The outer wall is comprised

of an aluminum bumper, nextel and kevlar fabric and a pressure shell made of steel or aluminum depending on the requirement of radiation

Starship Payload:

Payload	Transfer	Landing
GNC & OBDH	✓	✓
RF Communication Antenna	Side mounted	Top mounted
Laser Communications	✓	✗
Crew deck	✓	✓
Washrooms, Social Area, Dining & Fitness, etc	✓	✓
Medical, Scientific & Recycling Bays	✓	✓
Sleeping Pods, Solar Panels, Batteries	✓	✓
Food, Water, O ₂ & CO ₂ bays	✓	✓
Cargo at the base	✓	Multiple levels
Kilopower Reactor	Experimental payload	Main source for ground operations
EVA & Airlock at the base level	✓	✓
Cargo & Passenger lift to ground	✗	✓
Interlevel lift	✗	✓

TABLE II: Transfer starship payload details

V. POWER

Power generation is achieved through high-efficiency tripo-junction GaAs solar panels and one Kilopower nuclear reactor. The decision to include a nuclear reactor, both on board the transfer and the landing spacecraft, is motivated by the increased redundancy and reliability compared to relying on a single power generation system (solar), as well as demonstrating the reactors capabilities and safety features. For the lander, it would also serve as the main source of power for critical systems and in-situ resource utilization (ISRU), providing continuous reliable long-term power for many years. In contrast, solar panels degrade and do not provide power at night.

In addition to the power requirements for all the systems of the Starship, a security factor of 1.5 was adopted to ensure that the production of power stays well above the nominal operational values. A brief reliability study has been provided to show that, considering the limited duration of the mission (996 days), the reliability of the power systems remains well within acceptable limits.

The power required for the pumping of the ammonia and overall heat dissipation, will be maximum in the proximity of Earth, and will decrease as the ship gets closer to Mars. This will have a positive impact on the overall power consumption.

Kilopower Reactor

As an experimental payload and contingency backup, the reactor fuel in the Kilopower reactor is an alloy consisting of 93% uranium-235 and 7% molybdenum. The reactor core is surrounded by a beryllium oxide reflector that helps prevent neutrons from escaping and ensures that the nuclear chain reaction continues. Additionally, the reflector reduces gamma radiation emissions that could

potentially damage both the crew and on-board electronics [4]. The space-rated 10 kW Kilopower for Mars is expected to have a mass of 1500 kg in total (with a 226 kg core) and contain 43.7 kg of U_{235} , the heat generated by the fission is then carried to the Stirling converters, seen in figure 5, with passive sodium heat pipes [5].

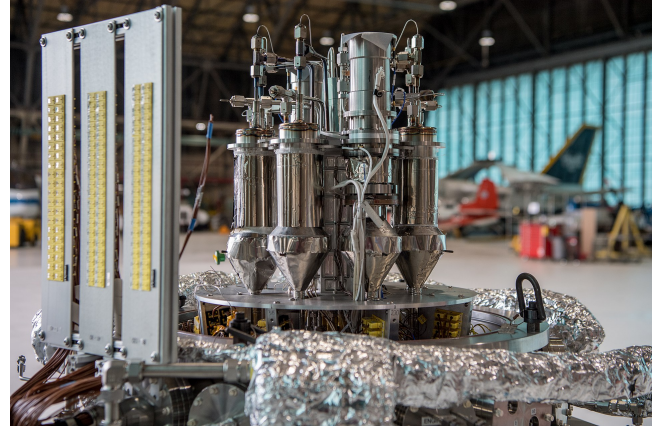


Fig. 5: Kilopower Reactor

Solar Panels

The solar flux varies significantly depending on the spacecraft's position relative to the Sun, ranging from 1361 W/m^2 at Earth to 590 W/m^2 at Mars. According to the spacecraft different subsystems consumptions the amount of power produced from the solar panels is around 25kW, while the reactor 10kW of electrical power. The choice of this panel is due to its high efficiency and the resistance to radiation degradation, while they must operate within 20 to 50 °C to maintain an optimal efficiency. The sizing was done by considering several factors, such as the degradation due to radiation and the lower solar flux around Mars. The solar panels are mounted on rotary joints, to help orientate them to an optimal position during the different phases of the transfer. This also serves to reduce their solar exposure, and therefore excess heat production, during periods of lower power consumption. Solar panels will experience efficiency degradation over time as a result of radiation, but the degradation rate will depend strongly on the solar cell technology and on the location of the spacecraft. With borosilicate glass panel coverings, this may be between 5-10% efficiency loss per year. Other glass coverings, such as fused silica and lead glasses, may reduce this efficiency loss to less than 1% per year [6]. Regardless, the power produced is well within, even at End of Mission, the power requirements.

Batteries

The Starship uses Lithium-ion polymer batteries to supply power during eclipse periods or particularly short high peak demands of energy. The choice of this kind of battery is due to their high specific power, high temperature range that will allow to reduce the amount of heat to dissipate.

As can be seen from table III the Nickel-Hydrogen batteries (used on the ISS) are suitable for long-duration missions, especially where eclipses are experienced daily.

Battery Type	Specifications
Lithium-Ion	Specific Energy: 180-220 Wh/kg Cycle Life: 400-1200 cycles Operating range (C°) : 10-45 Energy density: 250-693 Wh/L
Lithium Polymer	Specific Energy: 100-265 Wh/kg Cycle Life: 300-800 cycles Operating range (C°) : 50-70 Energy density: 250-670 Wh/L
Nickel-Hydrogen	Specific Energy: 55-75 Wh/kg Cycle Life: over 20000 cycles Operating range (C°) : -10-45 Energy density: 60 Wh/L

TABLE III: Comparison of Different Battery Technologies

However, the longevity exceeds the mission length, and the energy density is far lower than the other options. Between the Li-ion batteries or the Li-polymer ones, the polymeric one was chosen due to the high Specific Energy and range of operating temperature suiting the mission parameters.

The aim is for the solar panels to produce approximately 25kW, given by the power budget in table VI. The reactor, at its maximum, will produce 10kW of electric power. This requirement dictates the necessary solar panel area. For the chosen transfer, the minimum solar panel area required is 128.40 m², which is the area calculated while orbiting Mars. The amount of power produced while closer to Earth will be greater.

The spacecraft employs Li-polymer batteries to supply power during eclipse periods and during peaks of power consumption, without the necessity to increase the power output of the reactor. The energy required for battery storage takes into account the duration of the eclipse time around Earth, and other minor eclipses the ship might experience during the transfer (alignment of Earth-Sun-Mars). The energy required from the battery during eclipses is given by:

$$E_{battery} = P_{output} \cdot t_{eclipse}, \quad (1)$$

where $t_{eclipse}$ is the duration of eclipse (hours). Four hours of eclipse time was used as contingency in accordance to the Flight Design subteam's estimations. The total battery mass is estimated using:

$$M_{battery} = \frac{E_{battery}}{\text{energy density} \times \eta}. \quad (2)$$

For the duration of the transfer, the battery mass is estimated to around 210 kg, with a charging power required of about 13.8 kW. This supplies enough power to keep all systems on board operational.

Reliability of Power Systems

The reliability was calculated using Weibull and Gaussian distributions. Preliminary results were obtained, as seen in Figure 6. The whole duration of the mission, with some margin, is shown. The reliability of the systems is well within acceptable failure rate values. A parallel configuration for the pumps was chosen to reduce the risk of failures in the pumping system.

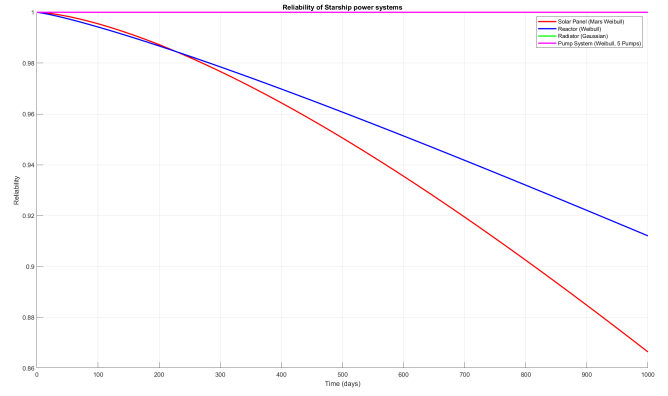


Fig. 6: Reliability model

VI. COMMUNICATION AND DATA HANDLING

Being a historic first manned deep space mission, communications are a crucial component in the design of both the space and ground segment. To ensure safety of the mission and crew, emphasis was put on a reliable and redundant communications system, with an onboard data handling segment to leave the opportunity for fully autonomous activities, e.g. in the case of communication outage. The requirements to the communications system can be found in IX. For each mission phase, the required reliable and high rate bandwidth in up- and downlink were defined, as can be found in 9.

The reliable channels are used primarily for speech communication with the astronauts (down-/uplink), as well as digital commands to the spacecraft and telemetry/engineering data from the spacecraft. Additionally, low definition video channels monitoring the astronauts activities are reserved. In addition to the reliable commanding, some bandwidth of the high rate channels is reserved for large command loads, for example updated procedures and software. This channel is expected to have a low relative duty factor. The remaining high rate uplink is allocated to high definition audio and video, used for communication with e.g. mission control, doctors and friends and family of the astronauts. Audio and video also occupy some downlink bandwidth, with the main portion of the bandwidth being used for scientific data gathered throughout the mission.

Due to the large differences in required bandwidth, link range and visibility between different mission phases, multiple approaches to communication are necessary. In the following sections, the individual system components are described in more detail. Link budgets detailing the feasibility of these concepts can be found in the appendix.

High Gain Antenna System

High Gain Antennas (HGAs), communicating in X-/or Ka-Band are the most commonly used method of high-rate communication with spacecraft. As the name implies, they feature a concentrated beam which allows for high data rates but requires accurate pointing of the antenna to a receiver. For the transfer vehicle, two deployable antennas on either side of the spacecraft are included in the design. This offers redundancy as well as the ability to

communicate with both lander vehicle and ground stations (see X). The lander vehicle has its HGA placed in the Level 1 of the payload area, to face upwards towards orbiting relay satellites (see VI). For the high gain system at maximum Mars-Earth distance (4×10^8 km), 60 Mbps downlink in the Ka-Band range with 100 W transmission power leaves a sufficient link margin of 4.5 dB (calculation done using Excel Link Budget tool found at [7]).

Low Gain Antenna System

Reliable communications during critical phases of the mission are handled via omnidirectional low gain antennas (LGA). They offer lower required pointing accuracy at the cost of lower received power due to their virtually isotropic beamshape. They are essential for all mission phases with large uncertainties in the spacecrafts orientation, including launches, landings and emergency situations. Four fully redundant antenna systems are located on the side of both lander and transfer vehicle. During launch and landing from and on Earth, they will be used to communicate with ground stations on Earth, whereas on Mars data will be transmitted via a relay system (see VI). [8].

Optical Communication System

The third communications system, found only on board the transfer vehicle, is an optical communication system. Laser-based optical communication systems have found success in Earth orbit applications [9] and, more recently, for deep-space purposes as demonstrated by the Deep Space Optical Communications (DSOC) payload onboard NASA's PSYCHE spacecraft [10]. Optical communications systems require a very high pointing accuracy but can provide high data rates at comparatively low power consumption. From DSOC's previous results, we expect between 8 and 250 Mbps of downlink capabilities, the former being at the maximum distance between Earth and Mars (roughly 4×10^8 km).

Areostationary Relay Satellite

During the stay on Mars' surface, the planet's rotation means that visibility between a given point on the surface and Earth cannot be guaranteed. Therefore, the use of an Areostationary Relay Satellite (ARS) is proposed. The satellite is placed in an equatorial circular orbit with an altitude of 17031.5 km [11], allowing its footprint to be at rest w.r.t. Mars' surface. This means it can have constant communications with the lander vehicle and relay data between ground stations and lander. The ARS adds additional complexity to the system and is estimated to cost roughly 1B USD (based on the cost of the Mars Reconnaissance Orbiter [12] plus margins). However, since it is beneficial in the long-term to establish permanent Mars colonies and significantly reduces risk and complexity of specifically the lander vehicle, it is considered a viable option. A visualization of the system including the ARS can be found in 7.

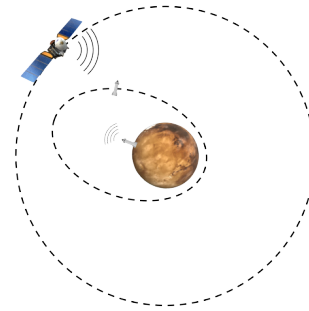


Fig. 7: Mars Communications System including ARS and Transfer Vehicle

Ground Stations

The basis of the ground based communications system is the Deep Space Network (DSN), "NASA's international array of giant radio antennas that supports interplanetary spacecraft missions" [13]. NASA states uplink capabilities up to 20 Mbps "are planned for complete implementation over the next few years" [14], which suffice for our uplink requirements. Additionally, it can handle laser communications, as shown with the aforementioned DSOC experiment [15].

Onboard Data Handling

To offload bandwidth from the communications systems, multiple offline data handling solutions are proposed in addition to the existing data handling for orbital maneuvers, GNC, etc.

In the downlink direction, scientific data taken during transit and on Mars shall be partially stored on the spacecraft, with the latter data being transferred back to the Transfer Vehicle while docked before the return trip. The ISS, which can be seen as comparable regarding generation of data (due to a similar crew size and focus on scientific utilization), produces roughly 28 TB of data a day [16], which is not feasible for our mission with either live transmission or onboard storage. We therefore assume this can be reduced to roughly 1 TB of storage needed per mission day, by increasing the amount of onboard processing and thereby reducing the amount of data needed in storage.

To reduce the amount of uplink traffic, a specialized large language model tailored to the mission, which has access to telemetry regarding spacecraft and astronauts, shall be available offline for assistance during nominal operation. Currently, models such as those based on Meta's LLaMA [17] require roughly 60 GB of RAM and solid state storage for offline use, so we assume a 128 GB RAM system and a similar amount of solid state storage needed to host a more advanced system to be developed in the 5 years leading up to the mission. It is to be noted that this generative artificial intelligence only serves to assist and reduce the need for thorough procedures for non-critical situations, and may not make or execute decisions without the astronauts' explicit permission. It does not alleviate the need for thorough procedures for all phases of the mission, which shall be on board of both vehicles from the launch

onward, with the possibility of updating throughout the mission. An overview of a potential onboard data handling system based on off the shelf components can be found in IV.

Component	Relevant Specifications	Units	Total Mass	Total Power
Payload Computer[18]	4 GB RAM	32	41.44	560 W
Rad-Tolerant SSDs [19]	4.5 TB	250	187.5 kg	-

TABLE IV: OBDH System Overview

VII. THERMAL CONTROL

Thermal regulation is necessary to dissipate excess heat generated by solar radiation, nuclear reactor, electronic equipment, and onboard systems.

A. Active Thermal Control

Since the the range of temperatures will vary significantly during the trip to Mars, solely having a passive thermal control system is not sufficient. Therefore, an active system has been developed to work in parallel when required similar to the ISS [20]. Due to the space environment, excess heat can only be dissipated through radiation. therefore a radiator system has been adopted. The system is a closed loop cycle, with liquid ammonia circulating through the radiator panels. where the mass flow varies according to the operational conditions. The active system comprises of:

- Pumps, to pump the liquid ammonia around the radiators, through pipes.
- Heat pipes to move around the higher temperature ammonia, toward the radiators from the solar panels and reactor. "Heat pipes are two-phase flow heat transfer devices where a process of liquid to vapor and vice versa circulates between evaporator and condenser with high effective thermal conductivity" [21].

The Stefan-Boltzmann law governs radiative cooling:

$$Q_{rad} = \sigma \epsilon A_{rad} T^4 \quad (3)$$

where σ is the Stefan-Boltzmann constant, ϵ is the emissivity of the radiator, A_{rad} is the radiator area, and T is the operating temperature.

The spacecraft uses an ammonia-based heat exchanger for thermal regulation. The mass flow rate of ammonia required to dissipate the absorbed heat is calculated as:

$$\dot{m}_{NH_3} = \frac{Q}{c_p \Delta T} \quad (4)$$

where c_p is the specific heat capacity of ammonia and ΔT is the temperature difference.

Based on calculations provided by the Life Support and Thermal System Subteam, the amount of ammonia necessary for cooling the solar panels will decrease as the ships distance to the Sun increases, to the point where the amount of fluid circulation needed to keep the surfaces at operational temperature will be negligible. The heat to dissipate for maintaining a temperature of 50°C is 118 032.85 W. The estimated radiator mass is 1200 kg, obtained with an area of 100 m², this results in an ammonia mass flow rate required for cooling of 1 kg/s around Earth.

VIII. RISK ANALYSIS

Risk analysis is a critical component in assessing and managing potential threats in any project, particularly in complex missions such as space exploration. The provided framework evaluates risks based on three key parameters: Risk Mitigation Scale, Likelihood Scale, and Consequence Scale. These elements help quantify and systematically address risks to ensure mission safety and success. The risk analysis table presented here provides a structured assessment of potential hazards in critical subsystems related to space missions. It systematically evaluates risks based on their Likelihood (L), Consequence (C), and Risk Level (L × C) while also listing corresponding Mitigation Strategies and their effectiveness through a Mitigation Level (ML, 1-5) which helps in reducing the overall risks for the mission in the early stage by developing a tabular data.

The table consists of several key columns:

- Risk ID: A unique identifier for each risk (e.g., R-01, R-02).
- System/Subsystem: The affected spacecraft system, such as radiation shielding, life support, power systems, and structural integrity.
- Risk Description: A concise explanation of the specific hazard, such as oxygen generation failure, micrometeoroid damage, or psychological stress in crew members.
- Likelihood (L): A rating from 1 (Very Low) to 5 (Very High), indicating the probability of the risk occurring.
- Consequence (C): A rating from 1 (Negligible) to 5 (Critical), representing the severity of impact if the risk materializes.
- Risk Level (L × C): The overall risk score, which is the product of Likelihood and Consequence values. Higher scores indicate more severe risks that require immediate mitigation.
- Mitigation Strategies: The specific countermeasures implemented to reduce the risk, such as multi-layered shielding, redundant cooling loops, or virtual reality entertainment for crew mental health.
- Mitigation Level (ML, 1-5): A rating showing how effectively the mitigation strategies reduce risk, with 5 being the strongest mitigation and 1 being ineffective.

Risk Mitigation Scale: The Risk Mitigation Scale ranges from 1 to 5, where 1 represents an unmitigated risk with no countermeasures in place, and 5 signifies a strongly mitigated risk with minimal residual threat. The scale categorizes risks as: Strongly Mitigated (5): Risks that have been effectively controlled, ensuring only minor residual threats remain. Well Mitigated (4): Risks that are largely managed but may still pose minor concerns. Partially Mitigated (3): Risks that persist despite countermeasures, requiring further attention. Weakly Mitigated (2): Limited success in risk reduction, necessitating additional strategies. Not Mitigated (1): No effective measures implemented, posing a significant threat.

Likelihood Scale (L): The Likelihood Scale determines the probability of a risk occurring, ranging from

Risk ID	System/Subsystem	Risk Description	Likelihood (L)	Consequence (C)	Risk Level (L × C)	Mitigation Strategies	Mitigation Level (ML) (1-5)
R-01	Radiation Shielding	High radiation exposure from cosmic rays and solar storms	5	5	25	Multi-layered shielding, water shielding, deployable storm shelters	4
R-02	Micrometeoroid & Space Debris Protection	Impact damage from high-velocity space debris	3	5	15	Reinforced outer shell, Whipple shield, real-time debris tracking & maneuverability	4
R-03	Solar Panels & Power System	Dust accumulation on solar panels reducing efficiency	4	4	16	Dust-resistant coatings, electrostatic cleaning, alternative power sources (RTGs, nuclear)	3
R-04	Nuclear Reactor Systems	Reactor cooling failure leading to overheating	2	5	10	Redundant cooling loops, passive heat rejection, automated shutdown protocols	4
R-05	Storage & Recycling Systems	Insufficient resource recycling causing supply depletion	3	4	12	High-efficiency closed-loop life support, backup consumables, redundancy in storage	4
R-06	Life Science Systems (ECLSS)	Oxygen generation system failure	3	5	15	Redundant oxygen supply, backup electrolysis units, algae-based O ₂ regeneration	4
R-07	Water Recycling & Filtration	Contaminant buildup in closed-loop water system	3	4	12	Multi-stage filtration, onboard water reserves, real-time contamination sensors	4
R-08	Thermal Control Systems	Inadequate temperature regulation leading to overheating/freezing	3	5	15	Active thermal management, phase change materials, redundant radiators	4
R-09	Structural Integrity	Stress fractures due to extreme temperature fluctuations and lander impact	2	4	8	Advanced composite materials, stress analysis, structural health monitoring sensors	5
R-10	Human Systems (Crew Ergonomics & Mental Health)	Crew isolation and psychological stress	4	4	16	AI-assisted support, virtual reality entertainment, scheduled social interactions	3
R-11	Emergency Evacuation & Abort System	No safe abort option in deep space/Martian surface	5	5	25	Redundant landers, emergency ascent vehicles, alternative return strategies	4
R-12	Landing System	Failure of precision landing technology leading to crash	3	5	15	Redundant landing thrusters, hazard detection systems, improved terrain mapping	4

Fig. 8: Risk Analysis Table summarizing key mission risks, their severity, and mitigation strategies.

1 (Very Low) to 5 (Very High): Very Low (1): The event is extremely unlikely. Low (2): The event has a small probability of occurring. Medium (3): The event has a moderate chance of happening. High (4): The event is likely to occur. Very High (5): The event is almost certain to happen.

Consequence Scale (C): The Consequence Scale evaluates the severity of an event's impact, ranging from 1 (Negligible) to 5 (Critical): Negligible (1): Minimal or no impact. Low (2): Small but manageable consequences. Medium (3): Moderate impact that requires corrective action. High (4): Significant damage, requiring extensive mitigation efforts. Critical (5): Severe impact, possibly mission-threatening

Mitigating risks associated with long-duration space travel is a crucial aspect of designing a human-rated Mars Starship. The mission faces multiple hazards, including radiation exposure, structural failures, life support malfunctions, and psychological challenges. To address these risks, a combination of engineering solutions, redundancy systems, and crew training has been incorporated into the spacecraft's design. One of the greatest threats in interplanetary space is radiation from cosmic rays and solar storms. The Starship has been designed with multi-layered shielding, using lightweight yet effective materials such as polyethylene composites, water-lined walls, and metal laminates. Water reservoirs and cargo storage have been strategically placed around the crew quarters to provide additional passive radiation protection. In case of solar storms, the spacecraft includes deployable storm shelters, where astronauts can take refuge in heavily shielded compartments until radiation levels return to safe limits.

IX. BUDGETS

Budgeting is a critical aspect of mission planning, ensuring that all necessary systems and infrastructure are

accounted for within a feasible financial framework. The budget for the Mars Starship mission encompasses development, manufacturing, power systems, communication, crew habitation, and supporting infrastructure, all essential for enabling a 969-sol round-trip mission to Mars.

The financial estimation includes major cost drivers such as Starship development and manufacturing, which accounts for the largest share of the budget at \$5.2 billion. Power generation systems, including a nuclear reactor (\$645 million) and solar panels with radiators (\$312 million), ensure continuous energy availability. The communication and onboard data systems (\$400 million) support both low- and high-frequency communications, as well as optical relays for high-bandwidth data transfer. Additionally, the aerostationary relay orbiter (\$1 billion) will provide uninterrupted communication between the Mars surface and Earth.

Crew habitation costs are also considered, including crew area and lift systems (\$71 million) to support astronaut living and working environments. Overall, the total estimated budget stands at approximately \$7.63 billion, covering all necessary components for mission success while allowing for contingencies and cost adjustments.

System	Mass [Tons]
Radiation Shielding	44
Nuclear Reactor	1.5
Solar Panels + Radiators	5.2
Communications + Onboard Data	0.9
Payload	29
Total	80.6
With 20 % margin	96.7

TABLE V: Mass distribution of different systems

Subsystem	Power (kW)
Life Support	7
Batteries charge	0-14
Pumps	6
Communication	1
OBDH	2
Total (Nominal)	16-30
Total (Peak, $\times 1.5$)	24

TABLE VI: Power budget estimation

System	Cost [USD]
Nuclear Reactor	645 Mio.
Solar Panels + Radiators	312 Mio.
Communications + Onboard Data	400 Mio.
Crew Area and Lift	71 Mio.
Starship Development and Manufacturing	5200 Mio.
Areostationary Relay Orbiter	1000 Mio.
Total	7628 Mio.

TABLE VII: Financial budget rough estimation

X. OFF-NOMINAL SCENARIOS

Communications Outage

While the ARS provides a good solution to the issue of communications from Mars' surface, it also results in a reliance on a complicated system, with repair or replacement during the mission infeasible. In case the ARS becomes unavailable during the on-surface phase of the mission, multiple redundancies must be available to maintain, at least, communication on the reliable channels for the duration of the astronaut's stay on Mars. To satisfy this, the use of other existing orbiters, such as those of the Mars Relay Network, as well as the transfer vehicle are proposed. Its LGAs and HGAs are available and have capabilities to communicate with Earth even at the maximum Earth-Mars distance. Since it has antenna systems on opposing sides of the spacecraft, it can communicate with Mars and Earth simultaneously. Additionally, spacecraft such as the Mars Reconnaissance Orbiter have been used to relay data in the past and can be expected to be used for this purpose as well, as they also provide capabilities to communicate in Ka-Band [22].

Since these backup solutions are expected to have lower bandwidth at more infrequent periods, the crews training, on board procedures and artificial intelligence solutions will be used extensively to ensure the safety of the mission.

Solar Power Failure

In the event of a partial or complete failure of the solar panels—whether due to mechanical failure or other unforeseen causes—the spacecraft can rely entirely on the Kilopower reactor to maintain all necessary life support systems and communication relay. The spacecraft is also designed for extravehicular activity (EVA), allowing crew members to perform external repairs, with spare parts and pre-mission training in place for solar panel maintenance/repair. If repairs are not possible, the reactor will

serve as the final contingency, ensuring continuous power for the remainder of the spacecraft's mission.

XI. CONCLUSION

The Vehicle Design Group has successfully developed a comprehensive human-rated structural design for the Mars Starship, ensuring its capability to sustain a crewed mission for 969 sols on the Martian surface. The design process incorporated advanced engineering tools, including Fusion 360 for CAD modeling and Blender for high-fidelity rendering, to optimize the spacecraft's architecture for safety, efficiency, and long-term habitability.

Key structural modifications were made to accommodate life support systems, crew quarters, medical and scientific bays, and other critical infrastructure. The Environmental Control and Life Support System (ECLSS) was designed with redundant oxygen generation, CO scrubbing, and water recycling, ensuring a sustainable closed-loop environment. The crew deck layout was optimized for both operational efficiency and psychological well-being, integrating private quarters, communal spaces, and dedicated medical facilities to support the astronauts over extended durations.

To maintain constant and reliable communication, the vehicle incorporates a hybrid communication strategy, including low-frequency radio, high-frequency Ka-band, and optical laser-based communication for high-data-rate transmissions. The system is complemented by a relay satellite constellation to ensure continuous contact with Earth, even during planetary occultations.

Power generation and thermal management were also central to the design. A dual-source power system—featuring solar arrays and auxiliary batteries—ensures uninterrupted energy availability. Active heat dissipation systems, including radiators and heat exchangers, were implemented to manage thermal loads, particularly during atmospheric entry, surface operations, and power-intensive scientific activities.

A rigorous reliability analysis was conducted, evaluating failure scenarios, risk mitigation strategies, and budget constraints to enhance mission success probability. The integration of high-redundancy systems, predictive diagnostics, and real-time structural health monitoring ensures that potential failures can be detected and addressed proactively. Risk assessments were balanced with budgetary limitations, prioritizing mission-critical enhancements while maintaining cost-efficiency.

In conclusion, the Vehicle Design Group has delivered a structurally optimized, mission-ready spacecraft, capable of supporting human life on Mars while maintaining reliability, safety, and efficiency. The integration of advanced manufacturing techniques, cutting-edge communication solutions, and sustainable power systems ensures that the Mars Starship is well-equipped for long-duration interplanetary missions, bringing humanity one step closer to establishing a sustainable presence on the Red Planet.

REFERENCES

- [1] NASA, "Mars surface operations and eva planning for human exploration," NASA, Tech. Rep., 2022. [Online]. Available: <https://www.nasa.gov/directorates/spacex-mars-crew-eva>.
- [2] J. Chang, *Space exercise device*, GrabCAD, Sep. 2015. [Online]. Available: <https://grabcad.com/library/visacec-space-station-recreation-center-1>.
- [3] NASA and SpaceX, "Mars transit architecture using cargo starship," *NASA Technical Reports*, vol. 5, pp. 12–25, 2021. [Online]. Available: <https://www.nasa.gov/directorates/spacex-mars-cargo-starship>.
- [4] Wikipedia contributors, *Kilopower — Wikipedia, the free encyclopedia*, 2024. [Online]. Available: <https://en.wikipedia.org/w/index.php?title=Kilopower&oldid=1220805656>.
- [5] M. A. Gibson, D. I. Poston, and E. al., "Kilopower reactor using stirling technology (krusty) nuclear ground test results and lessons learned," NASA Glenn Research Center, Tech. Rep. NASA/TM—2018-219941, 2018. [Online]. Available: <https://ntrs.nasa.gov/api/citations/20180007389/downloads/20180007389.pdf>.
- [6] H. J. Fincannon, "Solar power generation," NASA Glenn Research Center, Tech. Rep., 2021. [Online]. Available: https://ntrs.nasa.gov/api/citations/20205002844/downloads/Solar%20Power%20Generation_2.pdf.
- [7] *Link Budget Calculation Tool*. [Online]. Available: <https://www.google.com/url?sa=t&source=web&rct=j&opi=89978449&url=https://www.quasonix.com/files/link-budget-template-worksheet.xlsx&ved=2ahUKEwjXpNubroyMAxXnFRAIHQNgbEQFnoECAoQAQ&usq=AOvVaw19pxnTOClmGIK2yfQIIm0A>.
- [8] "(PDF) Design and Development of High Gain Antenna for Mars Orbiter Mission," en, *ResearchGate*, Oct. 2024. [Online]. Available: https://www.researchgate.net/publication/268686105_Design_and_Development_of_High_Gain_Antenna_for_Mars_Orbiter_Mission (visited on 02/12/2025).
- [9] *Laser Communications Relay Demonstration (LCRD) - NASA*, en-US, Section: Laser Communications Relay. [Online]. Available: <https://www.nasa.gov/mission/laser-communications-relay/> (visited on 03/15/2025).
- [10] *NASA's Optical Comms Demo Transmits Data Over 140 Million Miles - NASA*, en-US, Section: Psyche Mission, Apr. 2024. [Online]. Available: <https://www.nasa.gov/missions/psyche-mission/nasas-optical-comms-demo-transmits-data-over-140-million-miles/> (visited on 03/15/2025).
- [11] L. Montabone, N. Heavens, J. L. Alvarelllos, et al., "Observing Mars from Areostationary Orbit: Benefits and Applications," en, *Bulletin of the AAS*, vol. 53, no. 4, Mar. 2021. DOI: 10.3847/25c2cfef.0cdca220. [Online]. Available: <https://baas.aas.org/pub/2021n4i281> (visited on 03/14/2025).
- [12] *Cost of the Mars Reconnaissance Orbiter*, en. [Online]. Available: <https://www.planetary.org/space-policy/cost-of-the-mars-reconnaissance-orbiter> (visited on 03/14/2025).
- [13] *What is the Deep Space Network? - NASA*. [Online]. Available: <https://www.nasa.gov/directorates/somd/space-communications-navigation-program/what-is-the-deep-space-network/> (visited on 03/14/2025).
- [14] "Deep Space Network Services Catalog." [Online]. Available: <https://deepspace.jpl.nasa.gov/files/820-100-H.pdf>.
- [15] *Pia26148 Dsns Experimental Hybrid Antenna Tracks Dsocs Laser Downlink*, en-US. [Online]. Available: <https://www.jpl.nasa.gov/images/pia26148-dsns-experimental-hybrid-antenna-tracks-dsocs-laser-downlink/> (visited on 03/15/2025).
- [16] *NASA Communications Network to Double Space Station Data Rates - NASA*, en-US, Section: Space Communications & Navigation Program, Dec. 2016. [Online]. Available: <https://www.nasa.gov/directorates/somd/space-communications-navigation-program/nasa-communications-network-to-double-space-station-data-rates/> (visited on 03/14/2025).
- [17] *Llama*, en. [Online]. Available: <https://www.llama.com/> (visited on 03/16/2025).
- [18] *Radiation Tolerant, Low Power Sierra GPU Single Board Computer*, en. [Online]. Available: <https://www.moog.com/content/dam/moog/literature/sdg/space/avionics/moog-sierra-gpu-single-board-computer-datasheet.pdf> (visited on 03/14/2025).
- [19] *4.5 TB 3U VPX Rad-Tolerant SSDR*, en. [Online]. Available: <https://www.mrcy.com/products/data-storage-and-transfer/space-data-recorders/model-rh304t> (visited on 03/14/2025).
- [20] NASA, "Active thermal control system (atcs) overview," National Aeronautics and Space Administration, Tech. Rep. 473486main, 2008. [Online]. Available: https://www.nasa.gov/wp-content/uploads/2021/02/473486main_iss_atcs_overview.pdf.
- [21] B. Zohuri and S. Moaveni, *Spacecraft Thermal Control Handbook, Volume I: Fundamental and Methods*. Cham: Springer, 2017, ISBN: 978-3-319-29840-5. [Online]. Available: <https://link.springer.com/book/10.1007/978-3-319-29841-2>.
- [22] *Mars Reconnaissance Orbiter Telecommunications*, en. [Online]. Available: https://descanso.jpl.nasa.gov/DPSummary/MRO_092106.pdf (visited on 03/14/2025).
- [23] SpaceX, *Starship user's guide*, Online, Mar. 2020. [Online]. Available: https://www.spacex.com/media/starship_users_guide_v1.pdf.
- [24] NASA, "Nasa spaceflight human-system standard," NASA, Tech. Rep., 2010. [Online]. Available: <https://www.nasa.gov/sites/default/files/atoms/files/nasa-spaceflight-human-system-standard-volume-2.pdf>.

APPENDIX

Notes on the division of work

The entire model was made in Fusion 360 by Sharvil and Marcus and rendered in Blender. Sketchbook was used as the drawing software for 2D images shown for initial idea generation. The communications and data handling aspect was handled by Linus and power and thermal was under Koohyar and Brian including the CAD for solar panels, radiators and calculations. Only 1 model was referenced which was the multiple exercise device. The mission patch is specific for this mission and not to be used for any other purpose.

Mission Phase	Launch	Mars Transfer	Mars Landing	On Mars	Return Transfer	Landing Earth
LINK RANGE (max) [km]	-	1,50E+08	1,50E+08	4,00E+08	2,24E+08	-
UPLINK						
	Total Data Rate	Total Data Rate	Total Data Rate	Total Data Rate	Total Data Rate	Total Data Rate
Speech	60	60	60	60	60	60
Digital Commands	10	10	10	10	10	10
SUM (LG)	70	70	70	70	70	70
Command loads	100	20	0	0	20	100
Audio	0	320	0	320	320	0
HDTV	0	10000	0	10000	10000	0
SUM (HG/LASER)	100	10340	0	10320	10340	100
DOWNLINK						
	Total Data Rate	Total Data Rate	Total Data Rate	Total Data Rate	Total Data Rate	Total Data Rate
Speech	60	60	60	60	60	60
Engineering Data	60	60	60	60	60	60
Video	600	300	600	600	600	600
SUM (LG)	660	360	660	660	660	660
Audio	0	320	0	320	320	0
HDTV	0	10000	0	10000	10000	0
Payload Images & Data	0	0	0	50000	0	0
SUM (HG/LASER)	0	10320	0	60320	10320	0

Fig. 9: Overview of the required bandwidth [in kb/s] for different mission phases.

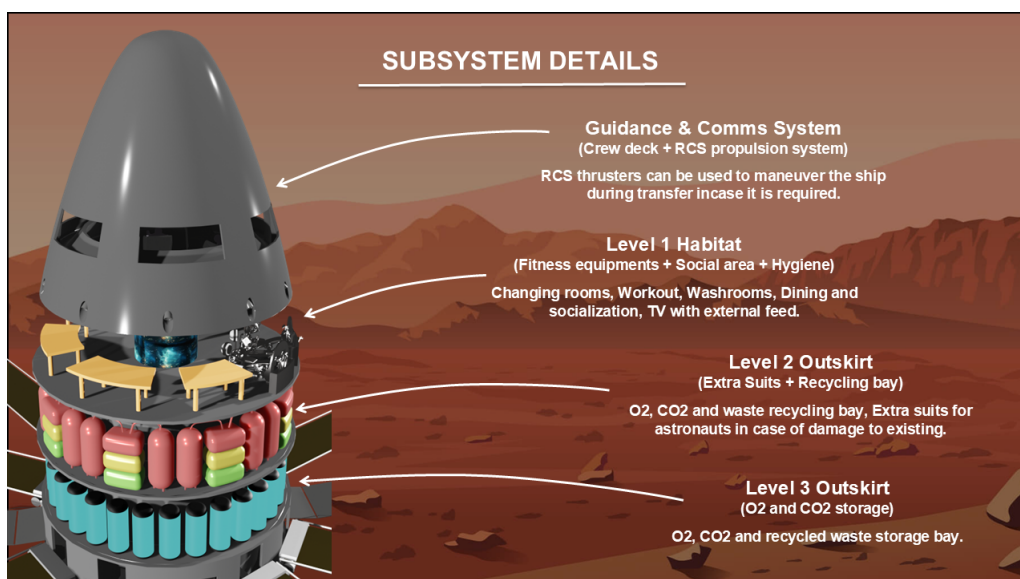


Fig. 10: Subsystem Level CAD Detail 1

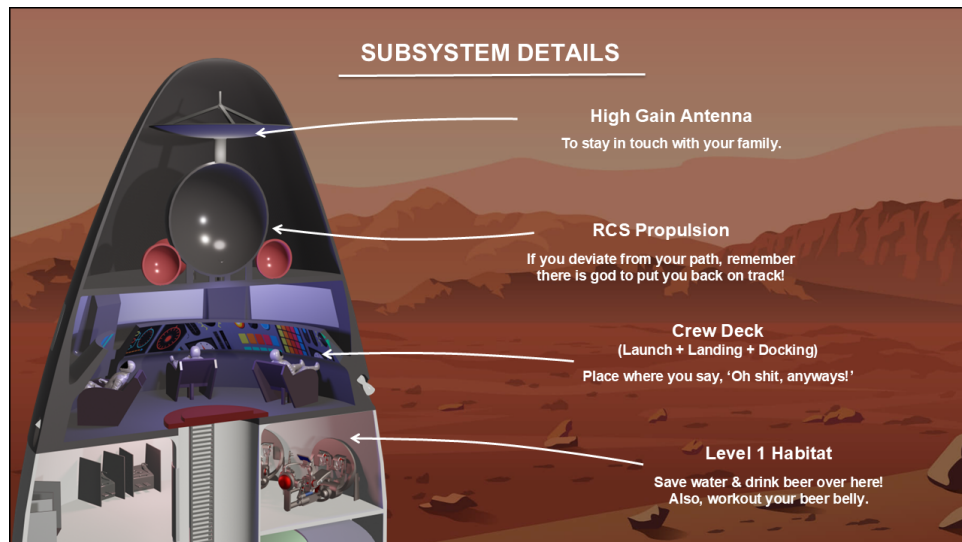


Fig. 11: Subsystem Level CAD Detail 2

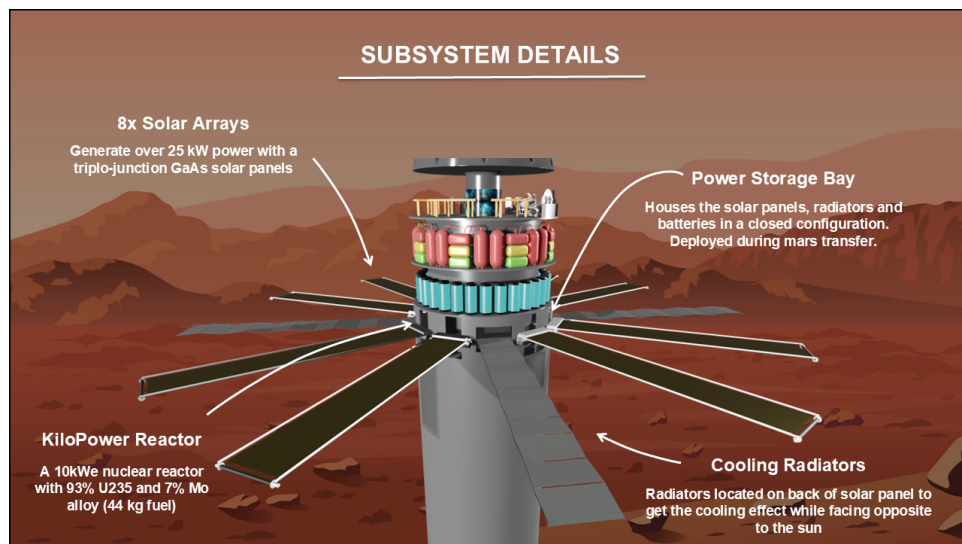


Fig. 12: Subsystem Level CAD Detail 3

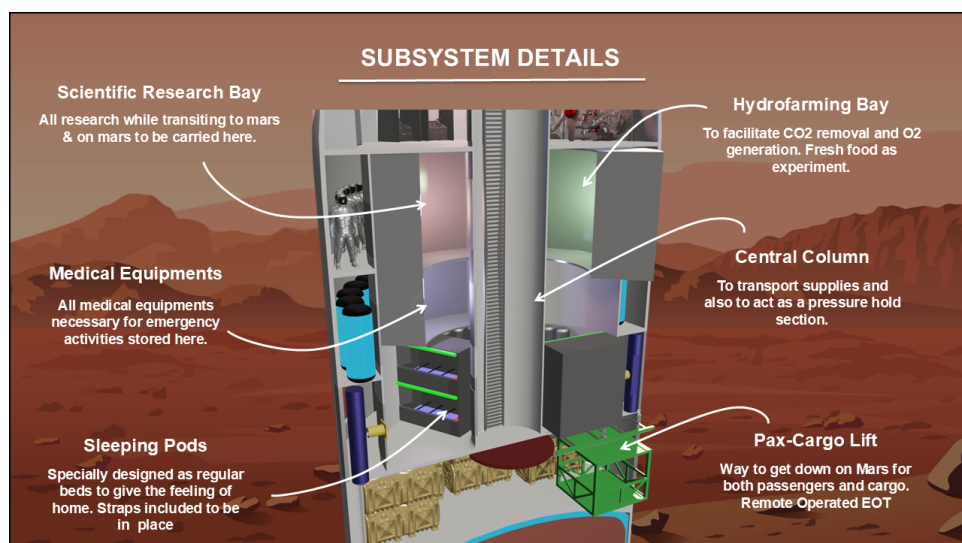


Fig. 13: Subsystem Level CAD Detail 4

ID	Description
VA-01	The propulsion system shall provide sufficient delta-V. This is the same as existing starship with some changes in RCS configurations. [23]
VA-02	The spacecraft shall be capable of autonomous navigation and course correction using onboard guidance and control systems.
VA-03	The entry, descent, and landing (EDL) system shall be capable of safely landing on Mars and executing ascent for return.
VA-04	The habitable volume shall provide at least 20 cubic meters per astronaut for living and working space. [24]
VA-05	The spacecraft shall include separate compartments for crew sleep, hygiene, medical care, and recreation.
VA-06	The Environmental Control and Life Support System (ECLSS) shall be capable of supporting six astronauts for the entire mission duration with air, water, and waste recycling.
VA-07	The power system shall provide at least 25 kW of continuous power using solar arrays and backup batteries.
VA-08	The thermal regulation system shall maintain an internal habitable temperature range between 23-26°C.
VA-09	The spacecraft shall have radiation shielding suited for crew protection throughout the mission.
VA-10	The communication system shall be well placed for radio and optical communication to be in continuous contact with Earth and relay satellites.
VA-11	The structural materials and subsystem mass shall be optimized to ensure the total payload mass does not exceed 100 tons.

TABLE VIII: L2 vehicle architecture requirements

ID	Requirement Description
COM-001	Disregarding non-avoidable communication blackouts, reliable communication channels between ground and space segments shall be constantly available during all phases of the mission.
COM-002	High-rate channels shall be available periodically throughout the mission.
COM-003	The bandwidth of both channels shall be sufficient for data loads during all mission phases.
COM-004	Both reliable and high-rate communication channels shall be sufficiently redundant to support the entire mission duration.
COM-005	The space system shall have sufficient data storage to store high-rate data in case of unavailability of the high-rate channels.
COM-006	The space system shall have offline capabilities to bridge non-avoidable reliable communication blackouts.

TABLE IX: L2 communications requirements

ID	Requirement Description
PTS-001	The power system shall consist of two separate power-generating sources, solar and nuclear, both individually capable of powering critical systems in case the other fails.
PTS-002	The power system shall include batteries capable of independently sustaining all critical spacecraft systems on board during eclipse periods.
PTS-003	The solar power system shall consist of solar panels capable of extending and retracting, with rotary joints for articulation to optimize power generation and minimize heat exposure during low-demand phases.
PTS-004	The overall thermal control system shall consist of a passive and an active system, working in parallel.

TABLE X: Power and thermal system requirements

Completed by:	System Level	Structure and Mass	Power	Communication	Heat/Cooling & Shielding	Payload	Financial
Week 5	Existing NCR design starship	NCR design	choice of power sources	link budget, commercial options	choice of cooling/heat shielding	types of payload	Initial cost estimates review
Week 6	Preliminary Design based on REQ (5.2.)	Base decks and layout			Shielding calcs [Sharvil]	non human payload	Preliminary estimates
Week 7	Finalize all main designs	Mass budget & changes	Finalize power budget and all info	Final comms design		final payload list	Final list
Week 8	Document everything	Final renderings & documentation					Changes and final budget
Week 9	Preliminary Result Presentation (27.2.)						
Week 10	Working in comments and critique						
Week 11	Documentation, finishing up, PROJECT DEADLINE (16.3.)						

Fig. 14: Weekly work timeline

